

# ANTHONY JARJOUR B.S.

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| EDUCATION       | <b>California State University, Fullerton</b> California, United States<br><i>B.S. Computer Science, Minor in Mathematics and Physics</i> 2021.08 - 2026.05 <ul style="list-style-type: none"><li>Extended degree by one year to formally pursue advanced physics and mathematics coursework and research in theoretical physics.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| RESEARCH        | <b>Research Assistant   Theoretical Physics – QFT</b> 2025.05 – Present <ul style="list-style-type: none"><li>Investigated thermodynamic properties of inequivalent Rindler vacua in Minkowski spacetime following Lochan et al. (2025), focusing on Bogoliubov transformations, vacuum inequivalence, and Planck-scale aspects of quantum field theory.</li><li>Built graduate-level theoretical foundation in General Relativity and Quantum Field Theory using Schutz (1985), Wald (1994), Ashtekar (1975), and Panangaden (1979), covering algebraic QFT, Green's functions and propagators, and Kähler structures.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| PRESENTATION    | <b>Bogoliubov Transformations Between Shifted Rindler Frames  </b> 2026.01<br><i>Joint Mathematics Meetings (JMM)</i> <ul style="list-style-type: none"><li>Presented at the 2026 Joint Mathematics Meetings an expanded study of Bogoliubov transformations between shifted Rindler frames, exploring how different quantization schemes yield distinct vacuum structures and physical interpretations.</li></ul> <b>Bogoliubov Transformations Between Shifted Rindler Frames  </b> 2025.10<br><i>Math Research Symposium</i> <ul style="list-style-type: none"><li>Delivered an in-depth exposition of Lochan &amp; Padmanabhan (2025), reproducing their derivation of Bogoliubov transformations between shifted Rindler frames and analyzing the resulting inequivalent vacuum structure in detail.</li></ul> <b>Data Science in Sport Analytics   ECS Innovation Expo</b> 2025.04 <ul style="list-style-type: none"><li>Cleaned and preprocessed soccer match data to develop machine learning models for predicting game outcomes, and presented application at an expo.</li></ul> |
| AWARDS & HONORS | <b>Dean's List   California State University, Fullerton</b> 2021.08 - 2026.05 <ul style="list-style-type: none"><li>Recognized for academic excellence for the Fall 2024, Spring 2025, Fall 2025, and Spring 2026 semesters</li></ul> <b>Funded MSRP Award   CSUF Department of Mathematics</b> 2025.05 <ul style="list-style-type: none"><li>Selected for the CSUF Department of Mathematics Math Summer Research Program to conduct 100 hours of paid undergraduate research.</li><li>Worked under faculty mentorship on a mathematics research project, culminating in a research poster and summary of findings.</li></ul> <b>3rd Place Award   CSUF CS Showcase</b> 2025.04 <ul style="list-style-type: none"><li>Awarded 3rd place for developing a machine learning model to predict soccer match outcomes, presented at the Computer Science Showcase at California State University, Fullerton.</li></ul>                                                                                                                                                                         |

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| TEACHING<br>EXPERIENCE | <b>Mathematical Circle Instructor - CSUF   Fullerton, California</b> Spring 2026 <ul style="list-style-type: none"> <li>• Lead a problem-solving session for middle- and high-school students in an open university outreach program, delivering a lecture and workshop-style activities.</li> <li>• Designed and curated challenging problem sets and guided solutions to strengthen mathematical reasoning, proof skills, and creativity.</li> <li>• Collaborated with faculty mentorship to select topics, refine lesson plans, and adapt instruction to varied student backgrounds and ability levels.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                        | <b>Undergraduate Tutor (Volunteer) - CSUF   Fullerton, California</b> Fall 2025 <ul style="list-style-type: none"> <li>• Mentored undergraduate students through homework and exam preparation in vector and tensor analysis.</li> <li>• Broke down complex derivations into approachable steps and emphasized geometric/physical interpretation alongside formalism.</li> <li>• Developed targeted practice problems and review notes to address recurring conceptual gaps.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| RELEVANT<br>COURSEWORK | <b>Completed</b><br>Computer Science: Compilers and Languages; Algorithm Engineering; Theory of Computation; Artificial Intelligence; Machine Learning<br>Mathematics: Vector and Tensor Analysis; Mathematical Probability; Partial Differential Equations; Differential Geometry; Quantum Field Theory; Intro to General Relativity<br>Physics: Thermodynamics, Kinetic Theory, and Statistical Physics; Classical Dynamics<br><br><b>Planned</b><br>Real Analysis I–II; Linear Algebra; Complex Analysis; Electromagnetic Theory I–II; Quantum Mechanics I–II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ACADEMIC<br>INTERESTS  | Theoretical Physics, Quantum Field Theory, General Relativity, Quantum Gravity, Computational Physics, Machine Learning for Physical Systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| PROJECTS               | <b>Large Language Model for Code Generation</b> 2024.09 – 2024.11 <ul style="list-style-type: none"> <li>• Technologies: Python, NLTK, Numpy, PyTorch, Pandas</li> <li>• Notes: Evaluated model output using unit tests to validate code correctness across multiple functional tasks, with a 93% pass rate.</li> </ul> <b>N-Body Problem Simulator</b> 2024.08 – 2024.10 <ul style="list-style-type: none"> <li>• Technologies: Python, Numpy, Matplotlib</li> <li>• Notes: Implemented Runge-Kutta integration to model two-body dynamics in a Newtonian gravitational system; visualized orbit evolution and field behavior.</li> <li>• Simulated gravitational fields using Newtonian mechanics with relativistic corrections</li> </ul> <b>Residual Neural Network Image Classifier</b> 2024.01 – 2024.05 <ul style="list-style-type: none"> <li>• Technologies: Python, PyTorch, NodeJS, Javascript, FastAPI, ScikitLearn, Pandas</li> <li>• Notes: Implemented an RNN based on K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," arXiv preprint arXiv:1512.03385, 2015, and achieved 95% validation accuracy with 0.25 loss, on bird species images</li> </ul> |

## SKILLS

### Programming Languages:

Python, C, C++, JavaScript, PHP, HTML, MATLAB, R

### Machine Learning / Data Science:

PyTorch, Torchvision, NumPy, Pandas, Scikit-learn, Matplotlib, Hugging Face Hub, Jupyter Notebooks

### Web / App Development:

React Native, Expo, Node.js, npm, FastAPI, Uvicorn, REST APIs

### Tools / Systems:

Git, GitHub, Linux/Ubuntu, VS Code, SQLite/MySQL-style database design

## INTERNSHIPS

### Cybersecurity Analyst - El Paso Water Utility | El Paso, Texas      2022.06 - 2022.08

- Designed and automated reports to visualize trends across multidimensional data streams, enhancing system performance and predictive diagnostics.

## SELECTED REFERENCES

- K. Lochan and T. Padmanabhan, "A Nested Sequence of Inequivalent Rindler Vacua: Universal Relic Thermality of Planckian Origin," *Classical and Quantum Gravity*, vol. 42, no. 3, 03LT01, 2025.
- B. F. Schutz, *A First Course in General Relativity*. Cambridge University Press, 1985.
- R. M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. University of Chicago Press, 1994.
- A. Ashtekar and A. Magnon, "Quantum Fields in Curved Space-Times," *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 346, no. 1646, pp. 375–394, 1975.
- P. Panangaden, "Positive and Negative Frequency Decompositions in Curved Spacetime," *Journal of Mathematical Physics*, vol. 20, no. 12, pp. 2506–2510, 1979.